



# GitHub Copilot Training: Mastering AI-Assisted Development

Welcome to our 2 Day comprehensive GitHub Copilot training session. Over the next two Days, we'll explore how this revolutionary AI assistant has transformed from a simple code completion tool into a full-fledged development partner that accelerates your workflow and enhances your coding experience.

By the end of this training, you'll understand how to leverage Copilot's advanced capabilities, configure it for optimal performance across multiple environments, and implement best practices that will make you and your team more productive and efficient.

**concentrix**<sup>™</sup>

# Day 1 Agenda



## Introduction to GitHub Copilot

Evolution, adoption, benefits, limitations, and future trends



## Setting Up Copilot

Installation, account setup, licensing, and configuration



## Copilot Fundamentals

Understanding AI suggestions, best practices, and hands-on exercises



## Q&A and Resources

Address questions and provide additional learning materials

We'll start with theoretical foundations to give you a solid understanding of how Copilot works, then move into practical applications that you can implement immediately in your development workflow.





# The Evolution of GitHub Copilot

1

2021

Initial release as code completion tool based on OpenAI Codex

2

2022-2023

Integration of Chat interface and expanded language support

3

2024

Introduction of Copilot Agent for autonomous PR creation and review

4

2025

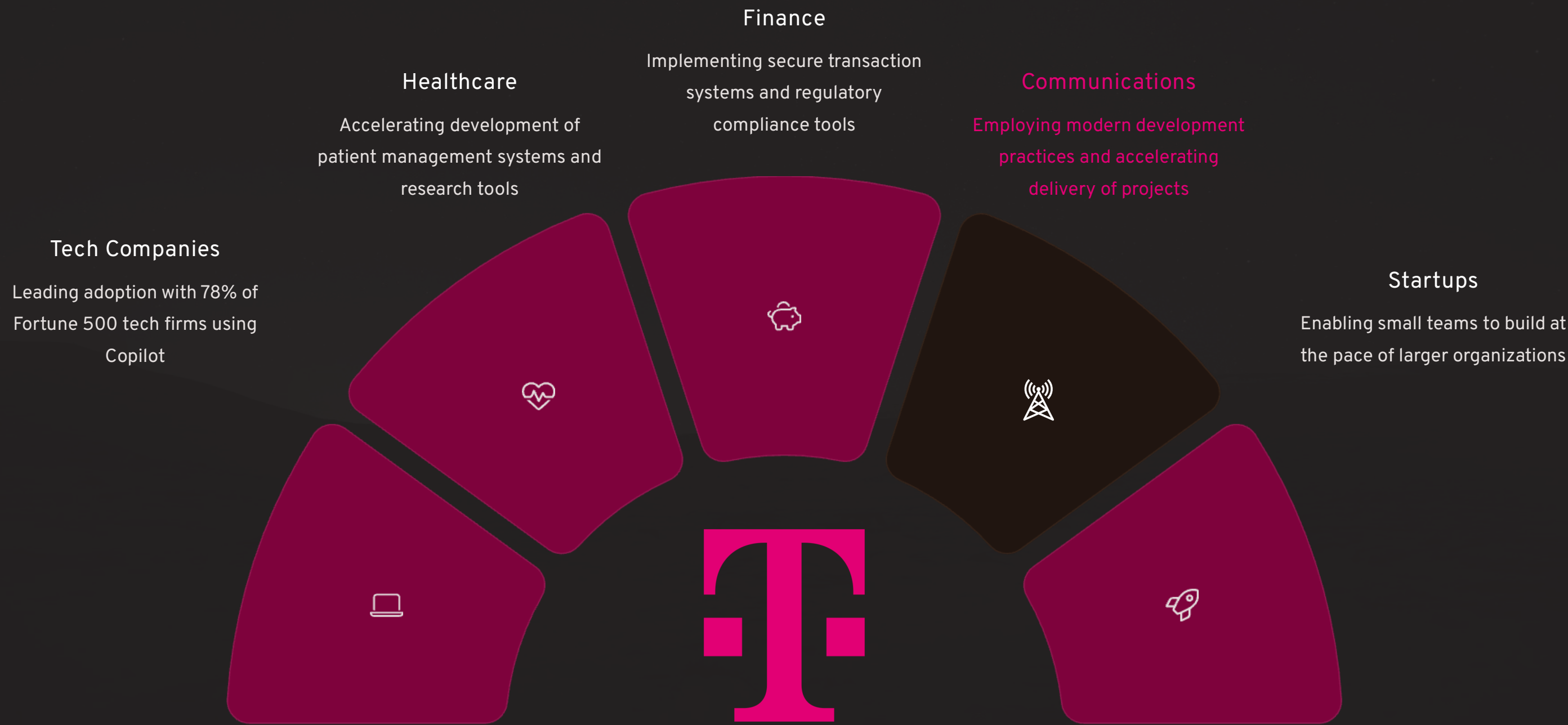
Full-featured autonomous coding partner with multimodal capabilities

GitHub Copilot has undergone a remarkable transformation since its inception. What began as a simple code completion tool has evolved into a sophisticated AI assistant that understands context across multiple files, generates entire functions, and can even create pull requests autonomously.

This evolution reflects broader advances in AI technology and GitHub's commitment to enhancing developer productivity. Today's Copilot doesn't just suggest code—it collaborates with you throughout the development lifecycle.



# Cross-Industry Adoption



GitHub Copilot has seen widespread adoption across industries, helping organizations of all sizes accelerate their development processes. With support for over 30 programming languages and integration with multiple IDEs, Copilot has become the preferred AI assistant for developers worldwide.



# Supported Languages and Environments

## Programming Languages

- JavaScript/TypeScript
- Python
- Java
- C/C++/C#
- Go
- Ruby
- PHP
- And 20+ more languages

## Integrated Development Environments

- VS Code
- JetBrains Suite
- Neovim
- Eclipse
- Xcode
- GitHub Codespaces
- Visual Studio

## File Types

- Code files
- Markdown
- JSON/YAML
- HTML/CSS
- SQL
- Documentation
- Test scripts



Documentation

GitHub Copilot's versatility is one of its greatest strengths. Whether you're writing front-end JavaScript, back-end Python, or low-level C++, Copilot provides intelligent suggestions tailored to your language and context. It seamlessly integrates with your preferred development environment, making it accessible regardless of your tech stack.





# Key Benefits of GitHub Copilot

## Enhanced Productivity

Studies show developers complete tasks up to 55% faster with Copilot, reducing time spent on repetitive coding patterns and boilerplate code.

## Improved Code Quality

Copilot suggests industry best practices and patterns, helping maintain consistent coding standards and reducing bugs by suggesting comprehensive error handling.

## Learning Acceleration

New developers can learn programming concepts by observing Copilot's suggestions, while experienced developers can explore unfamiliar libraries or languages more efficiently.

## Reduced Developer Fatigue

By handling routine coding tasks, Copilot allows developers to focus on solving complex problems and the creative aspects of software development.

These benefits translate to tangible business outcomes: faster time-to-market for new features, higher developer satisfaction, and more efficient use of engineering resources. Teams report spending more time on innovation and less on implementation details.

# Limitations and Considerations



## Requires Human Review

Copilot can generate incorrect, outdated, or insecure code. All suggestions must be carefully reviewed before implementation.



## Licensing Concerns

Generated code may inadvertently include patterns similar to open-source projects with restrictive licenses, raising potential intellectual property issues.



## Project-Specific Gaps

Copilot may lack understanding of proprietary frameworks, internal libraries, or company-specific coding standards without proper context.



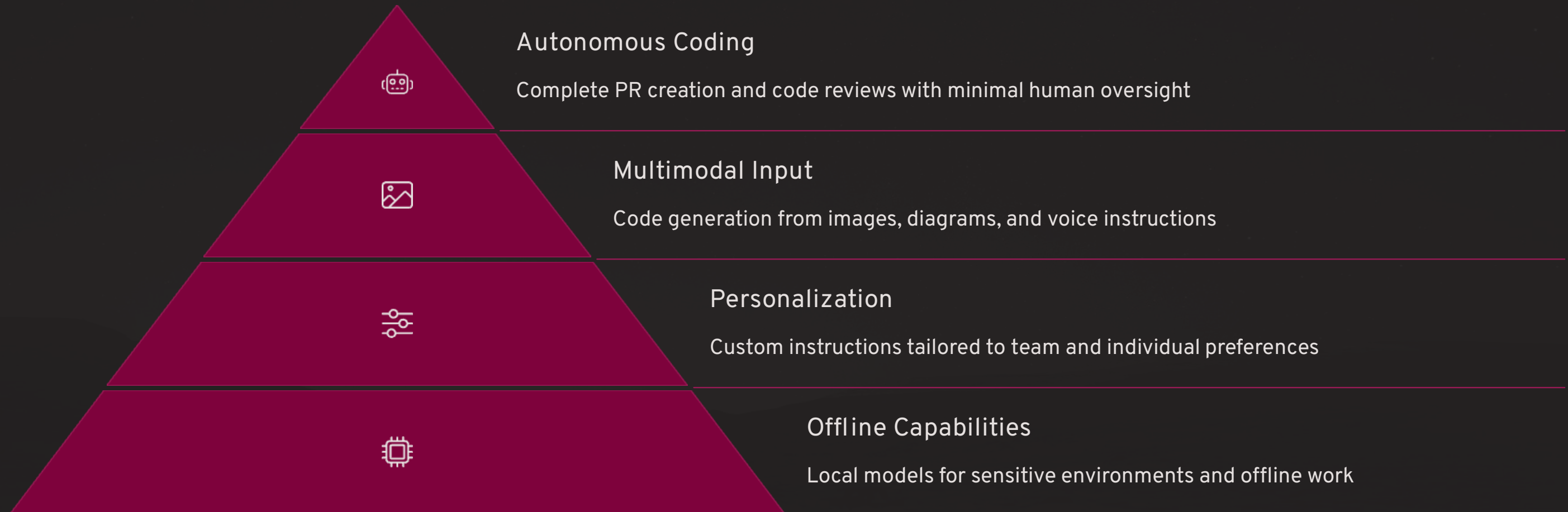
## Skill Dependency

Over-reliance may reduce developers' ability to write code without assistance, potentially affecting fundamental coding skills over time.

Being aware of these limitations is crucial for responsible use of AI coding assistants. Organizations should establish clear guidelines for Copilot usage, including when and how to validate suggestions, especially for security-sensitive or mission-critical code.



# Future Trends: Copilot in 2025



GitHub Copilot is rapidly evolving beyond simple code completion. The introduction of Copilot Agent now enables autonomous creation of pull requests based on issue descriptions, while new multimodal capabilities allow you to generate code from screenshots, diagrams, and even hand-drawn sketches.

Looking ahead, we can expect tighter integration with development workflows, more sophisticated understanding of project context, and the ability to explain its suggestions in natural language conversations. Voice and AR/VR coding experiences are also on the horizon, fundamentally changing how we interact with our development environments.





# Installation Across Different Environments

## IDE Extension Installation

Install GitHub Copilot from your IDE's extension marketplace ([VS Code](#), [JetBrains](#), etc.) or using the command line for Neovim. Each IDE has a dedicated extension optimized for its environment.

## GitHub Authentication

Sign in with your GitHub account when prompted. The extension will verify your Copilot subscription status automatically and request appropriate permissions to function.

## Verify Installation

Test Copilot by creating a new file in your preferred language and writing a comment describing functionality. Copilot should begin offering suggestions as you type, confirming successful installation.

Installation is straightforward across all supported environments, typically taking less than five minutes. For organizations using managed development environments, administrators can deploy Copilot at scale using standard software distribution tools or IDE configuration management.

# Competitor Success with GitHub Copilot





# INSTALLATION DEMO

- vs code
- IntelliJ





# Sub Copilot Training Installation (VS Code)

ntrix



# 5 MIN BREAK OUT ROOM

## INSTALL GITHUB COPILOT IN YOUR IDE



# POLLING...





# Licensing Options

| License Type     | Target Users        | Features                                          | Cost (2025)     |
|------------------|---------------------|---------------------------------------------------|-----------------|
| Individual       | Solo developers     | All core features, personal use only              | \$12/month      |
| Business         | Teams and companies | Adds team admin, usage reports, security policies | \$25/user/month |
| Enterprise (TMO) | Large organizations | Advanced security, compliance, custom deployment  | Contact sales   |
| Education        | Students, teachers  | Full access for academic purposes                 | Free            |
| Open Source      | Maintainers         | Full access for verified maintainers              | Free            |

GitHub offers flexible licensing options to accommodate different user needs. Individual licenses are perfect for freelancers and personal projects, while Business licenses add the management features needed for professional teams. Enterprise licenses include additional security features, compliance tools, and custom deployment options.

Educational institutions can register for free access, and verified open-source maintainers also qualify for complimentary licenses. Volume discounts are available for organizations purchasing multiple seats.



# Privacy Controls and Data Usage (Org)



## Telemetry Controls

Configure what data GitHub collects about your Copilot usage. Options range from full telemetry to minimal required data for service operation.



## Suggestion Privacy

Control whether your code context is used to improve the Copilot model. Disable to prevent your code from being used for training.



## Public Code Filter

Enable filters to reduce suggestions that closely match public code repositories, addressing potential licensing concerns.



## Enterprise Isolation

Enterprise plans offer isolated instances that never share data outside your organization's boundary.

GitHub has significantly enhanced Copilot's privacy controls in response to user feedback and regulatory requirements. You now have granular control over what data is collected, how your code is used, and whether suggestions can include matches from public repositories.

For organizations working with sensitive intellectual property or regulated data, Enterprise licenses provide additional isolation and compliance features to meet strict security requirements.

# Prompt Engineering

- Prompt engineering is vital in AI-assisted programming because it directly influences the quality of code suggestions generated by AI models like GitHub Copilot. By crafting precise and well-structured prompts, developers can guide the AI to understand their intent and context, leading to more accurate and useful code suggestions.
- Effective prompt engineering enhances productivity by reducing the time spent on writing boilerplate code and allowing developers to focus on complex tasks. It also ensures consistency and adherence to coding standards across the team.
- Moreover, clear and specific prompts help address limitations such as licensing concerns and project-specific gaps, ensuring the AI-generated code is reliable and secure.
- In summary, prompt engineering maximizes the benefits of AI-assisted programming, leading to increased efficiency, improved code quality, and a streamlined development process.







# Planning Your Approach – (KEY TAKEAWAYS)



## Create a Development Plan

Outline the structure and components before coding begins.



## Establish Clear Rules

Define coding standards and architectural guidelines.



## Limit Conversation Sessions

Keep AI interactions focused on specific tasks to maintain context.



## Focus on One Feature

Develop complete features sequentially rather than jumping between tasks.



# Demo Project





# BREAK OUT ROOMS

(Teams of 5) Try to reproduce the demo



30 MIN





# POLLING...



# Q&A



End Day 1